

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

Total Marks: 65

ANSWER KEY (Version 1)

Time: 2h 30m

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Ans	Correct Option	Explanation
1	D	undefined	A number has a multiplicative inverse only if $1/a$ is defined. Since $1/0$ is undefined, 0 has no multiplicative inverse .
2	B	the index of the radical	By definition $b^{m/n} = (\sqrt[n]{b})^m$, so the denominator n of the exponent equals the index of the radical, and m is the power raised.
3	B	zero or positive, never negative	The mantissa is the decimal (fractional) part of a logarithm, found from the log table, and by definition it is always non-negative — the sign adjustment is absorbed entirely into the characteristic.
4	C	$\log_2(-4)$	Logarithm is defined only for positive arguments . Since no power of 2 gives a negative result, log of a negative number is undefined .
5	A	between 5 and 3	The reference position is placed right after the first non-zero digit. In 0.00532, that digit is 5, so the reference position falls between the 5 and the 3 .
6	A	multiplied by the same nonzero constant	HCF of P and Q equals HCF of mP and nQ for any nonzero constants m, n — we may freely scale by constants (never by a variable) without changing the HCF.
7	A	$(x-5)(x-6)$	Need two numbers multiplying to +30 and adding to -11: -5 and -6 . Check: $(x-5)(x-6) = x^2 - 11x + 30$. ✓
8	A	$\pm(7a-6b)$	$49a^2 - 84ab + 36b^2 = (7a)^2 - 2(7a)(6b) + (6b)^2 = \mathbf{(7a-6b)^2}$, so the square root is $\pm(7a-6b)$.
9	A	{11, -3}	$ x-4 =7 \Rightarrow x-4=7$ or $x-4=-7 \Rightarrow \mathbf{x=11}$ or $\mathbf{x=-3}$.
10	B	$3(-2) > 5(-2)$	$3(-2)=-6$ and $5(-2)=-10$. Since $-6 > -10$, this is true. Multiplying an inequality by a negative number reverses its direction .
11	D	Fourth	(3, -5) has x positive and y negative , which is the defining sign pattern of the fourth quadrant .
12	C	12	Since P is the midpoint of RS, $PR = PS = 6$. So $RS = PR + PS = 6 + 6 = \mathbf{12}$.

SECTION B — Short Question Answers (any 11 × 3 = 33 Marks)

Q. 2(i) — Unit 1: Scientific Notation Application 3 marks

Momina walks at 3.5 km/h for 12 hours. Find the distance, in scientific notation.

Distance = speed × time = 3.5×12 1

Distance = 42 km 1

In scientific notation: $42 = \mathbf{4.2 \times 10^1}$ km 1

Q. 2(ii) — Unit 1: Real-Number Application (Temperature Scales) 3 marks

Convert -89.2°C into Fahrenheit and Kelvin using $F = (9/5)C + 32$ and $K = C + 273$.

$$F = (9/5)(-89.2) + 32 = -160.56 + 32 = -128.56^{\circ}\text{F} \quad 1.5$$

$$K = -89.2 + 273 = 183.8 \text{ K} \quad 1.5$$

Q. 2(iii) — Unit 1: Properties of Equality & Inequality 3 marks

(a) If $x+2=y$ and $y=2x-3$, then $x+2=2x-3$. (b) If $3<4$, then $-3>-4$.

(a) This uses the **Transitive Property of Equality**: if $a=b$ and $b=c$, then $a=c$ (here $a=x+2$, $b=y$, $c=2x-3$).
1.5

(b) This uses the **Multiplicative Property of Inequality** (for $c<0$): multiplying both sides of $3<4$ by -1 reverses the sign, giving $-3>-4$. 1.5

Q. 2(iv) — Unit 2: Antilogarithm 3 marks

Find the antilog of: (a) 2.4324 (b) 1.5890

(a) Characteristic = 2, mantissa = .4324. From the antilog table, mantissa .4324 $\rightarrow$ digits 2708. Characteristic 2 means 3 digits before the decimal point. 1.5

$$\text{antilog}(2.4324) \approx 270.8$$

(b) Characteristic = 1, mantissa = .5890. From the antilog table, mantissa .5890 $\rightarrow$ digits 3882. Characteristic 1 means 2 digits before the decimal point. 1.5

$$\text{antilog}(1.5890) \approx 38.82$$

Q. 2(v) — Unit 2: Domain of a Logarithm 3 marks

Check whether $\log_2(7-x)$ is defined for $x = 0, 1, 6, 7$.

(a) $x=0$: $7-0=7>0 \rightarrow$ **defined** ($\log_2 7$)

(b) $x=1$: $7-1=6>0 \rightarrow$ **defined** ($\log_2 6$)

(c) $x=6$: $7-6=1>0 \rightarrow$ **defined** ($\log_2 1=0$)

(d) $x=7$: $7-7=0 \rightarrow$ **undefined** (log of zero does not exist) 3 (0.75 each)

Q. 2(vi) — Unit 2: Change-of-Base Identity 3 marks

Verify $\log_c a \cdot \log_a b = \log_c b$ using $a=4$, $b=64$, $c=2$.

$$\log_2 4 = 2 \quad (\text{since } 2^2=4) \quad 1$$

$$\log_4 64 = 3 \quad (\text{since } 4^3=64) \quad 1$$

$$\text{LHS} = \log_2 4 \times \log_4 64 = 2 \times 3 = 6. \quad \text{RHS} = \log_2 64 = 6 \quad (\text{since } 2^6=64)$$

$$\text{LHS} = \text{RHS} = 6, \text{ so the identity is verified.} \quad 1$$

Q. 2(vii) — Unit 4: Sum/Difference of Cubes & Squares 3 marks

Factorize completely: (a) $27+512x^3$ (b) a^6-64

$$\begin{aligned} \text{(a) } 27+512x^3 &= 3^3+(8x)^3 = (3+8x)(3^2-3(8x)+(8x)^2) \\ &= (3+8x)(9-24x+64x^2) \end{aligned} \quad 1.5$$

$$\begin{aligned} \text{(b) } a^6-64 &= (a^3)^2-8^2 = (a^3-8)(a^3+8) = (a-2)(a^2+2a+4)(a+2)(a^2-2a+4) \\ &= (a-2)(a+2)(a^2+2a+4)(a^2-2a+4) \end{aligned} \quad 1.5$$

Q. 2(viii) — Unit 4: HCF by Factorization 3 marks

Find the HCF of $12x^2+x-1$ and $15x^2+8x+1$.

$$12x^2+x-1 = 12x^2+4x-3x-1 = 4x(3x+1)-1(3x+1) = (4x-1)(3x+1) \quad 1$$

$$15x^2+8x+1 = 15x^2+5x+3x+1 = 5x(3x+1)+1(3x+1) = (5x+1)(3x+1) \quad 1$$

$$\text{Common factor: } (3x+1) = \text{HCF} = (3x+1) \quad 1$$

Q. 2(ix) — Unit 4: Square Root by Factorization 3 marks

Find the square root of $25a^4-30a^3+9a^2$.

$$25a^4-30a^3+9a^2 = a^2(25a^2-30a+9) \quad 1$$

$$25a^2-30a+9 = (5a)^2-2(5a)(3)+3^2 = (5a-3)^2 \quad 1$$

$$\text{So the expression} = [a(5a-3)]^2 = \text{square root} = \pm a(5a-3) = \pm(5a^2-3a) \quad 1$$

Q. 2(x) — Unit 5: Linear Inequality Word Problem 3 marks

Two times a number decreased by 5 is $\geq$ the number increased by 8. Find the number.

$$\text{Let the number be } x. \text{ Then: } 2x - 5 \geq x + 8 \quad 1$$

$$2x - x \geq 8 + 5 \quad 1$$

$$x \geq 13 \quad 1$$

Q. 2(xi) — Unit 5: Checking Solutions of Inequalities 3 marks

(a) $5y-12>0$; $y=3$ (b) $5-2x>-4x+5$; $x=4$

$$(a) 5(3)-12 = 15-12 = 3 > 0 \Rightarrow \text{TRUE, } y=3 \text{ satisfies the inequality} \quad 1.5$$

$$(b) \text{LHS: } 5-2(4)=5-8=-3. \text{ RHS: } -4(4)+5=-16+5=-11. \text{ Is } -3>-11? \text{ Yes.} \quad 1$$

$$\text{TRUE, } x=4 \text{ satisfies the inequality} \quad 0.5$$

Q. 2(xii) — Unit 5: Absolute Value Equation 3 marks

Solve: $|5x-3| = |x+7|$

$$\text{Case 1: } 5x-3 = x+7 \Rightarrow 4x = 10 \Rightarrow x = 5/2 \quad 1.5$$

$$\text{Case 2: } 5x-3 = -(x+7) = -x-7 \Rightarrow 6x = -4 \Rightarrow x = -2/3 \quad 1.5$$

$$\text{Solution set} = \{5/2, -2/3\}$$

Q. 2(xiii) — Unit 7: Midpoint Formula (Decimals) 3 marks

Find the midpoint of $(1.4, -1.5)$ and $(2.6, 3.5)$.

$$\text{Midpoint} = ((1.4+2.6)/2, (-1.5+3.5)/2) \quad 1.5$$

$$= (4/2, 2/2) = (2, 1) \quad 1.5$$

Q. 2(xiv) — Unit 7: Missing Vertex of a Rectangle 3 marks

$A(1,2)$, $B(7,2)$, $C(7,8)$ are three vertices of rectangle ABCD. Find D.

$$AB \text{ is horizontal (} y=2, \text{ from } x=1 \text{ to } x=7\text{); } BC \text{ is vertical (} x=7, \text{ from } y=2 \text{ to } y=8\text{).} \quad 1$$

$$CD \text{ must be horizontal (} y=8, \text{ parallel to } AB\text{) and } DA \text{ must be vertical (} x=1, \text{ parallel to } BC\text{).} \quad 1$$

$$\text{So D has x-coordinate 1 (matching A) and y-coordinate 8 (matching C): } D = (1, 8) \quad 1$$

Q. 3(i) — Unit 1: Trichotomy Property & Ratio Word Problem 5 marks

(a) Verify Trichotomy for $a = -2\sqrt{3}$, $b = -5$. (b) Split Rs. 600,000 penalty in ratio 1:2:3 among Aatika, Bilal, Faryal.

(a) $-2\sqrt{3} \approx -2(1.732) = -3.464$. Compare with $b = -5$. 1

Since $-3.464 > -5$ (less negative lies further right on the number line), we have **$a > b$** . 1

Exactly one of $a < b$, $a = b$, $a > b$ holds (here $a > b$) — this confirms the **Trichotomy Property**.

(b) Total ratio parts = $1+2+3 = 6$. One part = $600,000 \div 6 = \text{Rs. } 100,000$. 2

Aatika = $1 \times 100,000 = \text{Rs. } 100,000$; Bilal = $2 \times 100,000 = \text{Rs. } 200,000$; Faryal = $3 \times 100,000 = \text{Rs. } 300,000$

2

Check: $100,000 + 200,000 + 300,000 = 600,000 \checkmark$

Q. 3(ii) — Unit 2: Laws of Logarithms & Digit Count 5 marks

Given $\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 5 = 0.6990$, $\log 7 = 0.8451$: (a) Evaluate $\log(28/45)$. (b) Find digits in 7^{40} .

(a) $\log(28/45) = \log 28 - \log 45 = \log(4 \times 7) - \log(9 \times 5)$ 0.5

$= (\log 4 + \log 7) - (\log 9 + \log 5) = (2\log 2 + \log 7) - (2\log 3 + \log 5)$ 1

$= [2(0.3010) + 0.8451] - [2(0.4771) + 0.6990]$ 1

$= [0.6020 + 0.8451] - [0.9542 + 0.6990] = 1.4471 - 1.6532$

$= -0.2061$ (negative, as expected since $28/45 < 1$) 0.5

(b) $\log(7^{40}) = 40 \times \log 7 = 40 \times 0.8451 = 33.804$ 1

Characteristic = 33 $\Rightarrow$ number of digits = characteristic + 1 = **34 digits** 1

Q. 3(iii) — Unit 4: Factorization Applied to a Cubical Tank 5 marks

Volume = $125x^3 - 150x^2 + 60x - 8 \text{ m}^3$. Find height, surface area, painting cost @ Rs.32/m².

$125x^3 - 150x^2 + 60x - 8 = (5x)^3 - 3(5x)^2(2) + 3(5x)(2)^2 - 2^3 = (5x-2)^3$ 1.5

(a) Since Volume = (side)³, the **height (side) = $(5x-2) \text{ m}$** 0.5

(b) Total surface area of a cube = $6(\text{side})^2 = 6(5x-2)^2 \text{ m}^2$ 1

(c) Cost = rate $\times$ area = $32 \times 6(5x-2)^2 = 192(5x-2)^2 \text{ Rupees}$ 2

Q. 3(iv) — Unit 5: Compound Inequality & Absolute Value Equation 5 marks

(a) $5x+10$ is less than -35 or greater than -5 . Find x . (b) Solve $|2y-5|=9$ and show on a number line.

(a) Case 1: $5x+10 < -35 \Rightarrow 5x < -45 \Rightarrow x < -9$ 1.5

Case 2: $5x+10 > -5 \Rightarrow 5x > -15 \Rightarrow x > -3$ 1

Solution: **$\{x \in \mathbf{R} : x < -9 \text{ or } x > -3\}$** 0.5

(b) $2y-5=9 \Rightarrow y=7$; $2y-5=-9 \Rightarrow y=-2$. Solution set = **$\{7, -2\}$** 1.5

Number line: hollow/filled dots plotted at -2 and 7 . 0.5

Q. 3(v) — Unit 7: Midpoint Quadrilateral (Varignon-type) **5 marks**

$ABCD$: $A(1,0)$, $B(5,2)$, $C(2,6)$, $D(-3,2)$. E,F,G,H = midpoints of AB,BC,CD,DA . Find E,F,G,H ; EF,FG,GH,HE ; identify $EFGH$.

(a) $E = \text{midpoint}(A,B) = ((1+5)/2, (0+2)/2) = (3, 1)$

$F = \text{midpoint}(B,C) = ((5+2)/2, (2+6)/2) = (7/2, 4)$

$G = \text{midpoint}(C,D) = ((2-3)/2, (6+2)/2) = (-1/2, 4)$

$H = \text{midpoint}(D,A) = ((-3+1)/2, (2+0)/2) = (-1, 1)$ **2**

(b) $EF = \sqrt{[(7/2-3)^2 + (4-1)^2]} = \sqrt{[1/4+9]} = \sqrt{(37/4)} = \sqrt{37/2} \approx 3.04$

$FG = \sqrt{[(-1/2-7/2)^2 + (4-4)^2]} = \sqrt{16} = 4$

$GH = \sqrt{[(-1+1/2)^2 + (1-4)^2]} = \sqrt{[1/4+9]} = \sqrt{37/2} \approx 3.04$

$HE = \sqrt{[(3+1)^2 + (1-1)^2]} = \sqrt{16} = 4$ **2**

(c) $EF = GH$ and $FG = HE \Rightarrow$ both pairs of opposite sides are equal $\Rightarrow$ **EFGH is a parallelogram.** **1**

Chapters Covered: Unit 1 (Real Numbers) | Unit 2 (Logarithms) | Unit 4 (Factorization & Algebraic Manipulation) | Unit 5 (Linear Equations & Inequalities) | Unit 7 (Coordinate Geometry) | Answer Key — Version 1

— End of Answer Key —